O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCOM GROWTH PROJECT\03 DELIVERABLES\ELV\F23 - FINAL TAILINGS HANDLING & STORAGE\2000-F23-SLD-EL-10211.DWG 020CT25 9:04AM HOTSEAT04

630A
4 POLE
G
3~
F23-GE-123
415V
400kVA

F23-SR-002
415V FINAL TAILS SWITCHROOM

SWR EARTH BAR

FEEDER FROM F23-SW-023
2000-F23-SLD-EL-10201
TO F23-SW-023
2000-F23-SLD-EL-10201

F23-SW-123
FINAL TAILS 415V EMERGENCY MOTOR CONTROL CENTRE

MCC EARTH BAR

NEUTRAL BAR
NOTE 1

AFD

100A
100A
SD
NOTE 3
2A
63A
2A
32A
1
6A
32A
6A
32A
ATS
M
AUTO
MCCB CHANGEOVER
SWITCH 3 POLE 630A

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

415V 3NE~ 50Hz 630A 65kA for 1sec

INC

CONTINUED ON SHT 2
2000-F23-SLD-EL-10212

02F-17	02F-09	02F-05	2F-01	1F-01	02F-13	01F-01	01F-05	01F-09	
250A	32A	32A	250A	160A	20A	20A	20A	20A	
2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A
90A	70A	70A	90A	100/1A	70A	90A	70A	70A	
250A 250A	32A 32A				EOL	EOL	EOL	EOL	

1

69kW
F23-DB-001
FINAL TAILS PUMPS
11kV / 415V SWITCHROOM
L&SP DB
2000-F23-SCM-EL-10205

ISL
10kW
F23UPS002-ISL
240VAC UPS
MAINTENANCE
BYPASS SWITCH
2000-F23-SCM-EL-10206

10kW
F23-DB-201
FINAL TAILS PUMPS
415V SWITCHROOM
240VAC UPS DB
2000-F23-TLD-EL-10101

UPS
10kW
F23-UPS-002
FINAL TAILS PUMPS
415V SWITCHROOM
240VAC UPS
2000-F23-TLD-EL-10101

E-SHUT
F23SR002-LCS
NOTE 12

F23-SR-001
11kV FINAL
TAILS PUMPS
SWITCHROOM
BC
6kW
F23-BC-001
FINAL TAILS PUMPS
11kV SWITCHROOM
48V CHARGER
2000-F23-TLD-EL-10052

10kW
F00-DB-401
EVENT POND
L&SP DB
2000-F00-SCM-EL-10213

69kW
F23-DB-301
FINAL TAILS AREA
L&SP DB
2000-F23-SCM-EL-10207

t°
M
3~
45kW
F23-AG-404
FINAL TAILS
TANK AGITATOR
2000-F23-SCM-EL-10208

M 3~	M 3~	M 3~	M 3~
0.18kW	0.18kW	0.18kW	0.18kW
F23-FN-466	F23-FN-467	F23-FN-468	F23-FN-469
FINAL TAILS PUMP 1 GEARBOX OIL AIR COOLING FAN	FINAL TAILS PUMP 2 GEARBOX OIL AIR COOLING FAN	FINAL TAILS PUMP 3 GEARBOX OIL AIR COOLING FAN	FINAL TAILS PUMP 4 GEARBOX OIL AIR COOLING FAN
2000-F23-SCM-EL-10210	2000-F23-SCM-EL-10211	2000-F23-SCM-EL-10212	2000-F23-SCM-EL-10213

NOTES

1. EARTH-NEUTRAL LINK INSTALLED IN UPSTREAM F23-SW-023 INCOMER TIER. SIGNAGE SHALL BE PLACED ON F23-SW-123 SWITCHBOARD AND ON GENERATOR READING "M.E.N. LINK LOCATED IN F23-SW-023 INCOMER TIER"
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
4. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
5. CIRCUIT BREAKER TO BE SUBSTITUTED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
6. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

PFR	PHASE FAILURE RELAY
/ >	EARTH LEAKAGE RELAY
xxxA / xxxA	ST TRIP SET POINT / CB SIZE
AFD	ARC FLASH DETECTION
ECR	EARTH CONTINUITY RELAY
1 E-SHUT	SWITCHROOM EMERGENCY SHUTDOWN

2000-F23-SLD-EL-10211_Rev1

F23-SW-123 415V emergency MCC - single line diagram sheet 1